

The Boomi logo is positioned in the top left corner of a decorative header. The header itself is a horizontal rectangle with a gradient from purple on the left to orange on the right, overlaid with abstract, wavy, concentric line patterns. The word "boomi" is written in a white, lowercase, sans-serif font.

boomi

Transforming Data with the Data Process Step



Table of Contents

About the Activities	4
Activity 1: Split Documents	5
Create Folders and Install Processes	6
Examine the Process and Data	6
Add a Data Process Step to Split Documents	9
Activity 2: Search and Replace	6
Add a Search/Replace Process Step	7
Activity 3: Combine the Data	10
Map the XML data to a Flat File	11
Add a Data Process Step to Combine the Documents	14
Activities Wrap-Up	17



About the Activities

To complete the Activities, log into the [Boomi Platform](#) with your Training Account and proceed with the instructions. We recommend completing the activities in order, as they build on each other.



Activity 1: Split Documents

NextGen Workforce Solutions receives patient data in a single XML file. However, the system they need to upload the data into requires each patient's information to be in separate records. To solve this, they use the Data Process Step, which parses the XML file into individual records by row. It ensures the data is organized and ready to be loaded into the system accurately and quickly.

In this activity, you will complete the following:

- Create Folders and Install Processes
- Add a Data Process step to Split Documents

Create Folders and Install Processes

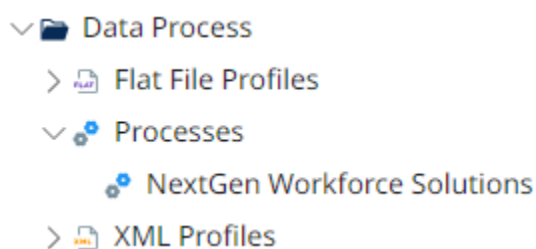
1. Create a folder named **Data Process** in the root folder.

Note: Not sure how to create a folder? Check out [Create a Folder](#).

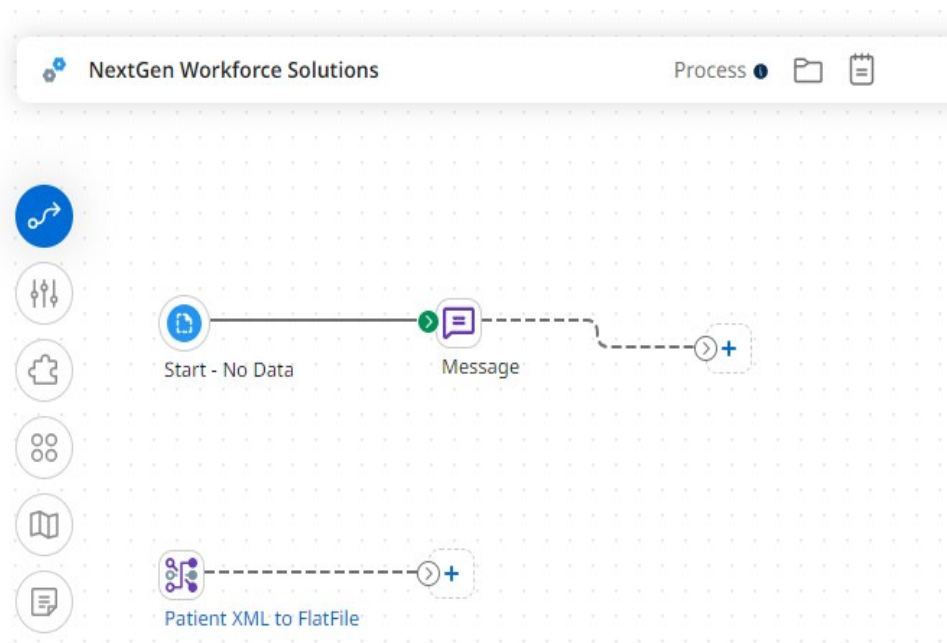
2. Install **NextGen Workforce Solutions** from the Process Library and put it into the Data Process folder.

Note: Not sure how to Install a Process from the Process Library? Check out [Install a Process from the Process Library](#).

Folder:



Process:



Examine the Message Step and Data

1. The process contains a Start, Message, and Map step. The Message step contains the XML data.

Message Step 1

To transform incoming document data into a free-form message using a combination of static and dynamic variables, use the Message step. The Message step is often used to construct email messages.

Display Name

Option

☐ Combine documents into a single message 1

Message

```
<Patients>
  <Patient>
    <ID>00!#01</ID>
    <Name>Alex Taylor</Name>
    <Address>456 Elm Street</Address>
    <City>Greenville</City>
    <State>TX</State>
    <Zip>75401</Zip>
  </Patient>
  <Patient>
    <ID>00!#02</ID>
    <Name>Ava Reynolds</Name>
    <Address>789 Oak Avenue</Address>
    <City>Brooklyn</City>
    <State>NY</State>
    <Zip>11201</Zip>
    <Count ry>USA</ Count ry>
  </ Patient>
  <Patient>
    <ID>00!#03</ID>
    <Name>Casey Morgan</Name>
    <Address>101 Pine Road</Address>
    <City>Seattle</City>
    <State>WA</State>
    <Zip>98101</Zip>
    <Count ry>USA</ Count ry>
  </ Patient>
</ Patients>
```

Variables

+

✎

✕

⬆

⬇

{1} Example variable

OK

Cancel

XML Data:

```
<Patients>
  <Patient>
    <ID>00!#01</ID>
    <Name>Alex Taylor</Name>
    <Address>456 Elm Street</Address>
    <City>Greenville</City>
    <State>TX</State>
    <Zip>75401</Zip>
    <Count ry>USA</ Count ry>
  </ Patient>
  <Patient>
    <ID>00!#02</ID>
    <Name>Ava Reynolds</Name>
    <Address>789 Oak Avenue</Address>
    <City>Brooklyn</City>
    <State>NY</State>
    <Zip>11201</Zip>
    <Count ry>USA</ Count ry>
  </ Patient>
  <Patient>
    <ID>00!#03</ID>
    <Name>Casey Morgan</Name>
    <Address>101 Pine Road</Address>
    <City>Seattle</City>
    <State>WA</State>
    <Zip>98101</Zip>
    <Count ry>USA</ Count ry>
  </ Patient>
</ Patients>
```

```

</ Patient >
< Patient >
  < ID>00!#04</ ID>
  < Name>Charlotte Hayes</ Name>
  < Address>202 Birch Lane</ Address>
  < City>Atlanta</ City>
  < State>GA</ State>
  < Zip>30303</ Zip>
  < Country>USA</ Country>
</ Patient >
< Patient >
  < ID>00!#05</ ID>
  < Name>Daniel Evans</ Name>
  < Address>303 Maple Drive</ Address>
  < City>Denver</ City>
  < State>CO</ State>
  < Zip>80202</ Zip>
  < Country>USA</ Country>
</ Patient >
< Patient >
  < ID>00!#06</ ID>
  < Name>Emily Carter</ Name>
  < Address>404 Cedar Street</ Address>
  < City>Chicago</ City>
  < State>IL</ State>
  < Zip>60601</ Zip>
  < Country>USA</ Country>
</ Patient >
< Patient >
  < ID>00!#07</ ID>
  < Name>Ethan Morgan</ Name>
  < Address>505 Redwood Avenue</ Address>
  < City>San Francisco</ City>
  < State>CA</ State>
  < Zip>94101</ Zip>
  < Country>USA</ Country>
</ Patient >
< Patient >
  < ID>00!#08</ ID>
  < Name>James Walker</ Name>
  < Address>606 Pine Street</ Address>
  < City>Los Angeles</ City>
  < State>CA</ State>
  < Zip>90001</ Zip>
  < Country>USA</ Country>
</ Patient >
< Patient >
  < ID>00!#09</ ID>
  < Name>Jordan Riley</ Name>
  < Address>707 Walnut Lane</ Address>
  < City>Austin</ City>
  < State>TX</ State>
  < Zip>73301</ Zip>
  < Country>USA</ Country>

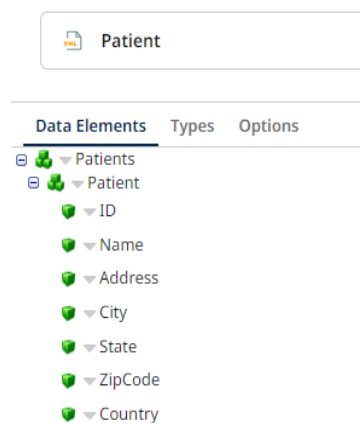
```

```

</Patient>
<Patient>
  <ID>00!#10</ID>
  <Name>Liam Fisher</Name>
  <Address>808 Willow Drive</Address>
  <City>Portland</City>
  <State>OR</State>
  <Zip>97201</Zip>
  <Country>USA</Country>
</Patient>
<Patient>
  <ID>00!#11</ID>
  <Name>Mason Coleman</Name>
  <Address>909 Fir Street</Address>
  <City>Miami</City>
  <State>FL</State>
  <Zip>33101</Zip>
  <Country>USA</Country>
</Patient>
<Patient>
  <ID>00!#12</ID>
  <Name>Olivia Brooks</Name>
  <Address>1001 Elm Avenue</Address>
  <City>New York</City>
  <State>NY</State>
  <Zip>10001</Zip>
  <Country>USA</Country>
</Patient>
<Patient>
  <ID>00!#13</ID>
  <Name>Sophia Bennett</Name>
  <Address>1102 Maple Street</Address>
  <City>Boston</City>
  <State>MA</State>
  <Zip>02108</Zip>
  <Country>USA</Country>
</Patient>
</Patients>

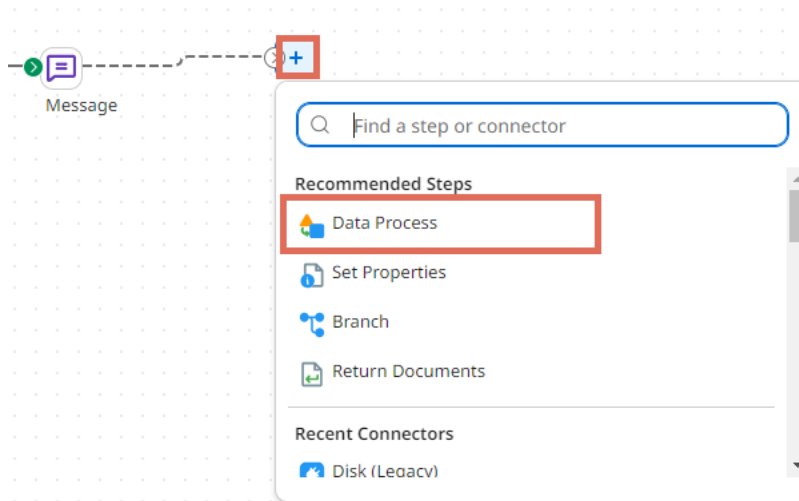
```

2. The process also includes an XML profile.



Add a Data Process Step to Split Documents

1. Select the + next to the Message to open the Data Process step.



2. Open the Data Process step and select the + below Data Process Properties.

Data Process Step ⓘ

To manipulate document data with your process, use the Data Process step. This step provides multiple options for data manipulation, like splitting or combining documents and zipping or unzipping data. You can add one or more processing steps, which execute according to their sequential order and operate on the data output from the previous processing step.

Display Name

Processing Steps

Data Process Properties ⓘ



OK

Cancel

3. Enter the following settings:

- Processing Step: **Split Documents**
- Profile Type: **XML**
- Batch Count: **0**
- Profile: **Patient**
- Split Element: **Patient**
- Then, select **OK**.

Data Process Step ⁱ

To manipulate document data with your process, use the Data Process step. This step provides multiple options for data manipulation, like splitting or combining documents and zipping or unzipping data. You can add one or more processing steps, which execute according to their sequential order and operate on the data output from the previous processing step.

Display Name

Processing Steps

Data Process Properties ⁱ

Split Documents

Processing Step ⁱ

Split Documents

Profile Type

XML

Split Options ⁱ

Batch Count

0

Profile

Patient

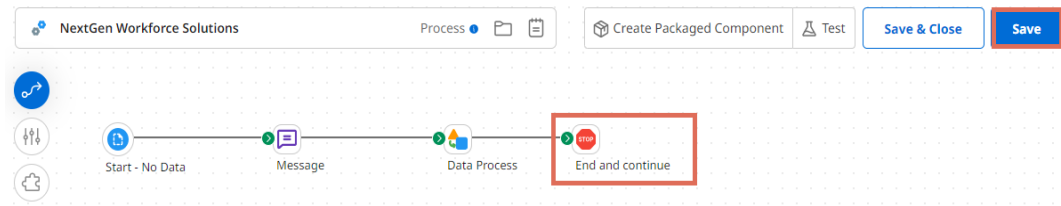
Split Element

Patient (Patients/Patient)

OK

Cancel

4. Add a **Stop** step to the end of your process and **Save** it.



3. Use the Test Atom Cloud to execute the Process. You should get 13 documents.

Unsure how to test your process? Check out [How to Test Your Process](#).



Activity 2: Search and Replace

While analyzing the data, a NextGen developer noticed that ID fields contained the characters '#!'. These characters could cause issues when loading the data into the output system. To fix this, the developer decided to remove these characters programmatically. While reviewing the Data Process step, he found a Search/Replace process step that could resolve this issue.

In this activity, you will complete the following:

- Add a Search/Replace process step

Add a Search/Replace Process Step

1. Open the Data Process step on the Process Canvas and click the + sign under Data Process Properties.

Data Process Step ⓘ

To manipulate document data with your process, use the Data Process step. This step provides multiple options for data manipulation, like splitting or combining documents and zipping or unzipping data. You can add one or more processing steps, which execute according to their sequential order and operate on the data output from the previous processing step.

Display Name

Processing Steps

Data Process Properties ⓘ



Split Documents

2. Enter the following settings:

- Processing Step: **Search/Replace**
- Text to Find: **!#**
- Replace With: (Blank out this field, no value)
- Search: Check "Incrementally by specified number of characters"
- Characters: **1024**

Then, select **OK**.

Data Process Step ⁱ

To manipulate document data with your process, use the Data Process step. This step provides multiple options for data manipulation, like splitting or combining documents and zipping or unzipping data. You can add one or more processing steps, which execute according to their sequential order and operate on the data output from the previous processing step.

Display Name

Processing Steps

Data Process Properties ⁱ

Split Documents

Search/Replace

Processing Step ⁱ

Text To Find ⁱ

Replace With

Search

Characters ⁱ

Search/Replace

!#

☐ Entire document at once (not recommended for large documents) ⁱ

☐ Incrementally line by line

☒ Incrementally by a specified number of characters

1024

OK

Cancel

3. **Save** the process. Use the **Test Atom Cloud** to execute the Process. Examine the output. You should see the ID containing only numbers.

Unsure how to test your process? Check out [How to Test Your Process](#).

Document Viewer

```
1 <Patients>
2 <Patient>
3 <ID>0001</ID>
4 <Name>Alex Taylor</Name>
5 <Address>456 Elm Street</Address>
6 <City>Greenville</City>
7 <State>TX</State>
8 <Zip>75401</Zip>
9 <Country>USA</Country>
10 </Patient>
11 </Patients>
```



Activity 3: Combine the Data

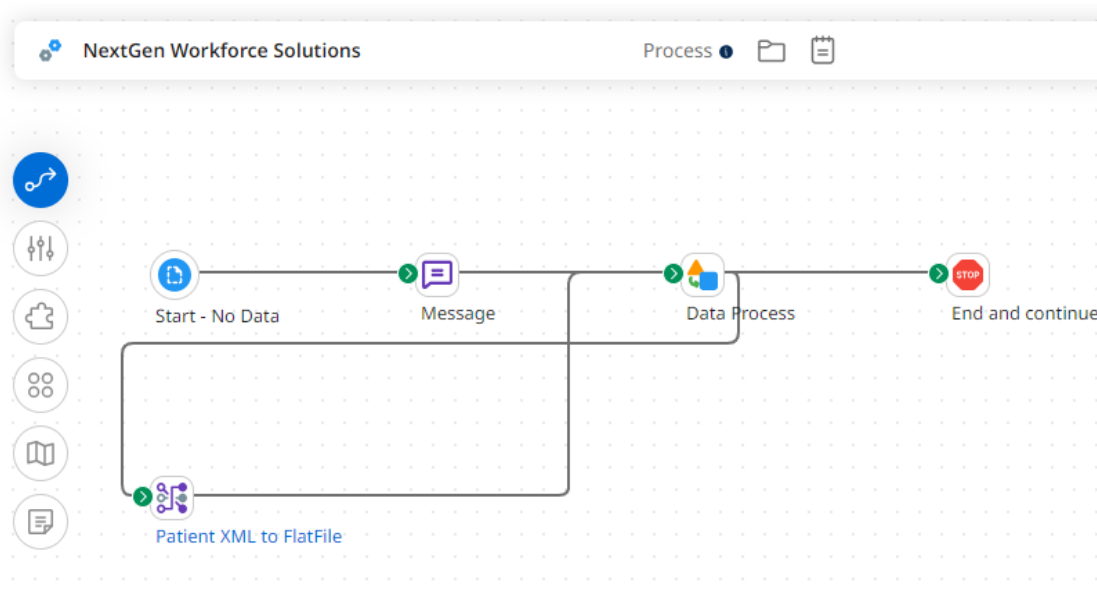
After mapping the data to a flat file, the NextGen developer learned the customer wanted the data in one file. He used the Combine Documents step in the Data Process to combine the data into one output file. When the process ran, the documents merged into one file, meeting the customer's needs.

In this activity, you will complete the following:

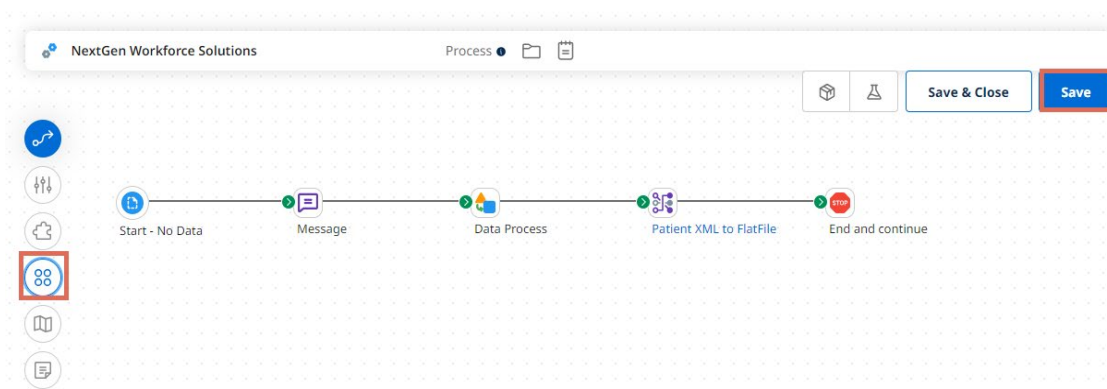
- Use the Map step to map the XML profile to the Flat File profile.
- Create a new Data Process step to combine the data into one file.

Map the XML data to a Flat File

1. From the Process Canvas, disconnect the **Data Process** from the **Stop** step and attach it to the **Map** step. Attach the Map to the Stop step shown in the image below.



2. Select the **Save** and **Arrange** buttons to clean up the Process Canvas. The process will look like the one below.



3. Select the **Map** step to open it, then select the **pencil** symbol to edit the map.

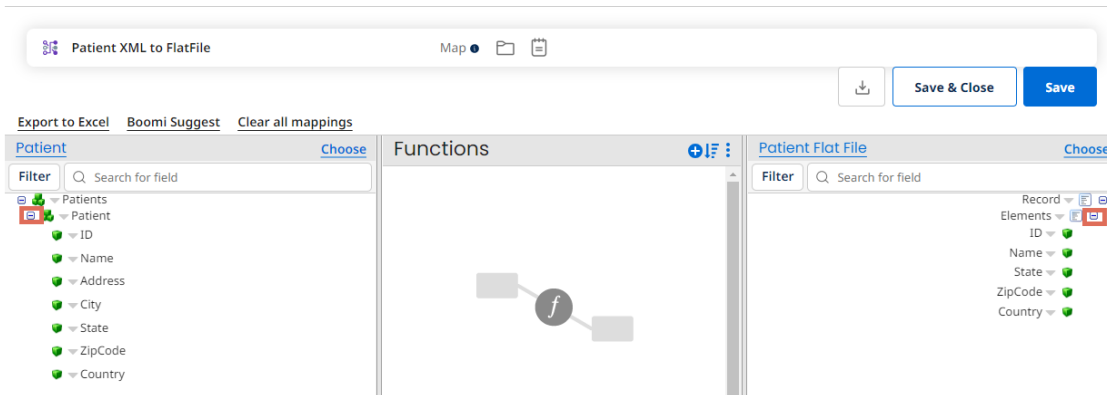
Map Step

Use maps to transform data from one format to another, or more specifically, from one profile to another. The Map step references a predefined Map component.

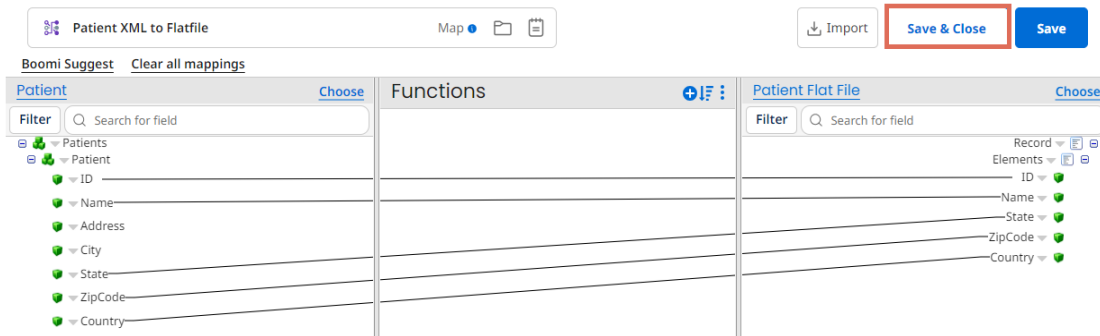
Display Name

Map  

4. Expand the Source and Destination profiles by selecting the highlighted boxes, as shown below.

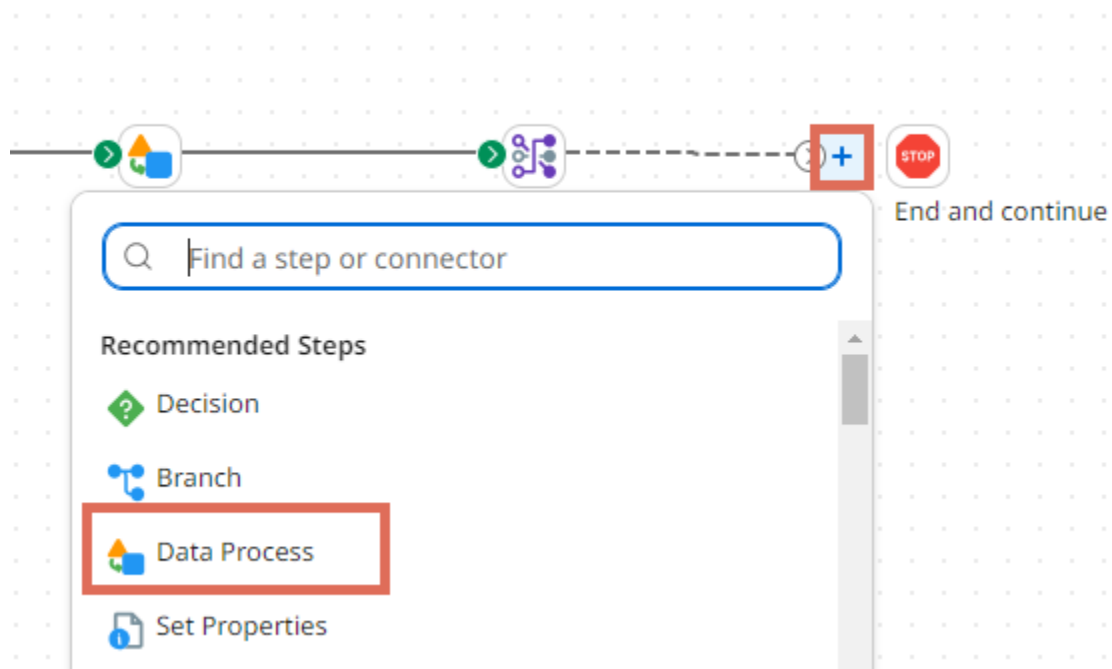


4. Manually map the fields so your map looks like the one below. Then, select **Save and Close** to save and leave the map, and **OK** to exit the Map Step.



Add a Data Process Step to Combine the Documents

1. From the Process Canvas, unhook the Map Step from the Stop step, and from the Map step, select the + symbol and **Data Process** step.



2. Select the **Data Process** step to open it, then select the **+** symbol to create new Data Process Properties. Enter the following settings:

- Processing Step: **Combine Documents**
- Profile Type: **Flat File**
- Headers Option: **Retain first line as Column Headers**

Then, select **OK**.

Data Process Step

To manipulate document data with your process, use the Data Process step. This step provides multiple options for data manipulation, like splitting or combining documents and zipping or unzipping data. You can add one or more processing steps, which execute according to their sequential order and operate on the data output from the previous processing step.

Display Name

Processing Steps

Data Process Properties ⓘ

+ × ↕ ⬇

Combine Documents

Processing Step ⓘ

Combine Documents ▼

Profile Type

Flat File ▼

Combine Options ⓘ

Free-Form Header

Free-Form Footer

Headers Option

☐ No Column Headers

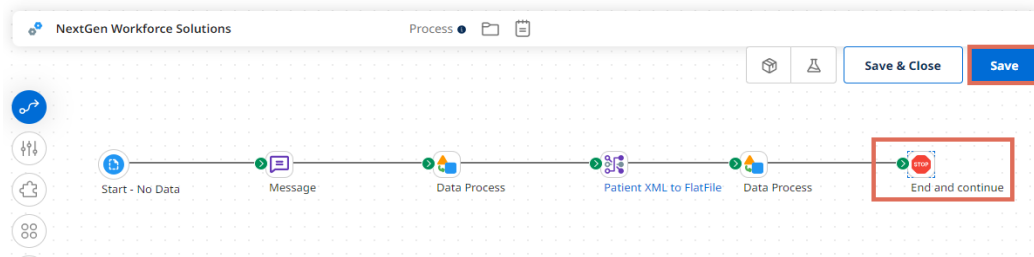
☐ Remove first line as Column Headers

☒ Retain first line as Column Headers

OK

Cancel

3. Connect the **Data Process** to the **Stop** step and **Save** the Process.



4. **Test** the process. Your output should look like the following:

Document Viewer

The Document Viewer displays a list of patient records with the following columns: ID, Name, State, ZipCode, and Country. The records are as follows:

ID	Name	State	ZipCode	Country
0001	Alex Taylor	TX		USA
0002	Ava Reynolds	NY		USA
0003	Casey Morgan	WA		USA
0004	Charlotte Hayes	GA		USA
0005	Daniel Evans	CO		USA
0006	Emily Carter	IL		USA
0007	Ethan Morgan	CA		USA
0008	James Walker	CA		USA
0009	Jordan Riley	TX		USA
0010	Liam Fisher	OR		USA
0011	Mason Coleman	FL		USA
0012	Olivia Brooks	NY		USA
0013	Sophia Bennett	MA		USA

Below the list, there is a link to [Download Original Document](#) and a note indicating the document size is 373 bytes. A 'Close Document Viewer' button is located at the bottom right.

Unsure how to test your process? Check out [How to Test Your Process](#).



Activities Wrap-Up

Congratulations! You have completed testing your **Understanding the Data Process Step** solution and all activities.



COPYRIGHTS AND TRADEMARKS

This guide contains proprietary information protected by copyright and/or other legal grounds. The software described in this guide is furnished under a software license, service agreement, or nondisclosure agreement. This software may be used or copied only under the terms of the applicable agreement. No part of this guide may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying and recording for any purpose without the written permission of Boomi, LP ("Boomi").

The information in this document is provided in connection with Boomi products. No license, express or implied, by estoppel or otherwise, to any intellectual property right is granted by this document or in connection with the sale of Boomi products and/or services. EXCEPT AS OUTLINED IN THE TERMS AND CONDITIONS AS SPECIFIED IN THE LICENSE AGREEMENT FOR THIS PRODUCT, BOOMI (TOGETHER WITH BOOMI TOPCO, LP, AND ITS DIRECT AND INDIRECT SUBSIDIARIES) ASSUMES NO LIABILITY WHATSOEVER AND DISCLAIMS ANY EXPRESS, IMPLIED, OR STATUTORY WARRANTIES RELATING TO ITS PRODUCTS INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, OR NON-INFRINGEMENT. IN NO EVENT SHALL BOOMI OR ITS AFFILIATES BE LIABLE FOR ANY DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE, SPECIAL, OR INCIDENTAL DAMAGES (INCLUDING, WITHOUT LIMITATION, DAMAGES FOR LOSS OF PROFITS, BUSINESS INTERRUPTION OR LOSS OF, DAMAGE TO, OR CORRUPTION OF DATA OR INFORMATION) ARISING OUT OF THE USE OR INABILITY TO USE THIS DOCUMENT OR INFORMATION, EVEN IF PARTIES HAVE BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. Boomi makes no representations or warranties concerning the accuracy or completeness of the contents of this document and reserves the right to make changes to specifications and product descriptions in its sole and absolute discretion at any time without notice. Boomi is not committed to updating the information in this document.

If you have any questions regarding your potential use of this material, contact:

Boomi, LP

Attn: LEGAL Dept.

Legal-Notice@boomi.com

With a copy to:

Boomi, LP – Legal Department, 1400 Liberty Ridge Drive, Chesterbrook, PA 19087

Trademarks

Copyright © 2025 Boomi, LP. All rights reserved. Boomi, the Boomi 'b' logo or astronaut logo, AtomSphere, Atom, Molecule, and AtomSphere Integration Cloud are trademarks of Boomi, LP, and/or its affiliates in the United States and/or other countries. Other trademarks and trade names may be used in this document to refer to either the entities claiming the marks and names or their products.